

A maternal vegetarian diet in pregnancy is associated with hypospadias. The ALSPAC Study Team. Avon Longitudinal Study of Pregnancy and Childhood.

OBJECTIVE: To investigate the possible role of the maternal diet, particularly vegetarianism and consumption of phytoestrogens, in the origin of hypospadias, which is reported to be increasing in prevalence. **SUBJECTS AND METHODS:** Detailed information was obtained prospectively from mothers, including previous obstetric history, lifestyle and dietary practices, using structured self-completed questionnaires during pregnancy. Previously recognized associations with environmental and parental factors were examined, focusing particularly on the hypothesized hormonal link. Multivariate logistic regression was used to identify independent associations. **RESULTS:** Of 7928 boys born to mothers taking part in the Avon Longitudinal Study of Pregnancy and Childhood, 51 hypospadias cases were identified. There were no significant differences in the proportion of hypospadias cases among mothers who smoked, consumed alcohol or for any aspect of their previous reproductive history (including the number of previous pregnancies, number of miscarriages, use of the contraceptive pill, time to conception and age at menarche). Significant differences were detected for some aspects of the maternal diet, i.e. vegetarianism and iron supplementation in the first half of pregnancy. Mothers who were vegetarian in pregnancy had an adjusted odds ratio (OR) of 4.99 (95% confidence interval, CI, 2.10-11.88) of giving birth to a boy with hypospadias, compared with omnivores who did not supplement their diet with iron. Omnivores who supplemented their diet with iron had an adjusted OR of 2.07 (95% CI, 1.00-4.32). The only other statistically significant association for hypospadias was with influenza in the first 3 months of pregnancy (adjusted OR 3.19, 95% CI 1.50-6.78). **CONCLUSION:** As vegetarians have a greater exposure to phytoestrogens than do omnivores, these results support the possibility that phytoestrogens have a deleterious effect on the developing male reproductive system.